ST231 Assignment

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Question 1

a.

The estimated coefficient for $\log(x)$ is 5.930 to 3.d.p. (*)

The value of this estimate does appear reasonable given the application context. Firstly, by using the volume of a sphere, the volume of a round diamond is approximately proportional to x^3. Since the mass of the diamond is proportional to the volume, we have the mass of a round diamond is approximately proportional to x^3.

Hence, if we have two round diamonds where diamond A is 10% heavier than diamond B then we expect the length of diamond A to be $(1.10^{\frac{1}{3}} - 1) \times 100 = 3.228\%$ (**) longer than the length of diamond B to 3.d.p.

By M1, we have

$$\log(\text{price}) = \beta_0 + \beta_1 \log(x).$$

Changing the subject for price we obtain

$$\text{price} = \exp(\beta_0) \times x^{\beta_1}.$$

Let x_A and x_B be the lengths of diamonds A and B respectively. Similarly, let price_A and price_B be the prices of diamonds A and B respectively.

Then, the

$$\begin{aligned}
\text{percentage difference in price} &= \frac{\text{price}_A - \text{price}_B}{\text{price}_B} \times 100 \\
&= \left(\frac{\exp(\beta_0) \times x_A^{\beta_1}}{\exp(\beta_0) \times x_B^{\beta_1}} - 1 \right) \times 100 \\
&= \left(\left(\frac{x_A}{x_B} \right)^{\beta_1} - 1 \right) \times 100 \\
&= ((1.10^{\frac{1}{3}})^{5.930} - 1) \times 100 && \text{by (*) and (**)} \\
&= 20.730\%
\end{aligned}$$

to 3.d.p. To the nearest integer, we expect diamond A to be approximately 21 % more expensive than diamond B, which aligns with the information shared by the jeweller.

b.

We consider the model M2 where for $j = 1, \ldots, 607$,

$$\mathbb{E}(\log(\text{Price}_j)) = \beta_0 + \beta_1 \log(x_j) + \beta_2 \log(y_j) + \beta_3 \log(z_j).$$

Using a similar approach to part (a), we obtain Volume $\propto xyz$. Hence, Price $\propto$ Weight^2 and thus Price $\propto (xyz)^2$. Taking logs on both sides yields $\log(\text{Price}) = 2\log(x) + 2\log(y) + 2\log(z)$ therefore we expect $\beta_i = 2$ for $i = 1, 2, 3$.

c.

```
Call:
lm(formula = log.price ~ log.x + log.y + log.z, data = dia)

Residuals:
    Min      1Q  Median      3Q     Max 
-0.4187 -0.1173 -0.0137  0.1282  0.3530 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  -1.7674     0.1524 -11.595  < 2e-16 ***
log.x        -0.1621     0.8102  -0.200    0.842    
log.y         4.3858     0.7800   5.623 2.87e-08 ***
log.z         1.7823     0.2544   7.006 6.59e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1579 on 603 degrees of freedom
Multiple R-squared:  0.9535,	Adjusted R-squared:  0.9533 
F-statistic:  4126 on 3 and 603 DF,  p-value: < 2.2e-16
```

Null and alternative hypothesis: $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 \neq 0$.

Firstly, considering a significance level of 5%, the p-value reported in the $\Pr(> |t|)$ column for $\log(x)$ in the model summary is $0.842 > 0.05$. Therefore, there is insufficient evidence to reject the null hypothesis.

d.

```
Analysis of Variance Table

Response: log.price
           Df  Sum Sq Mean Sq   F value    Pr(>F)    
log.x       1 306.762 306.762 12296.292 < 2.2e-16 ***
log.y       1   0.804   0.804    32.209 2.152e-08 ***
log.z       1   1.224   1.224    49.081 6.587e-12 ***
```

```
Residuals 603  15.043   0.025
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The **null and alternative hypotheses** for $\log(x)$ in the sequential ANOVA is $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 \neq 0$. We take a significance level of 1% where the p-value equals $\mathrm{P}(>|F|)$. In the $\mathrm{P}(>|F|)$ column of the output for $\log(x)$, the p-value is $<< 0.01$ hence there is insufficient evidence to reject the null hypothesis. Note this does not contradict our conclusion in part (c).

While the sequential ANOVA tests the marginal contribution of $\log(x)$ before $\log(y)$ and $\log(z)$ are added to the model, the t-tests in part (c) test if $\log(x)$ adds an explanatory influence after $\log(y)$ and $\log(z)$ are added to the model.

However, a small discrepancy does arise. As x, y, and z are highly correlated, once $\log(y)$ and $\log(z)$ are added to the model $\log(x)$ contributes little additional information to the model.

e.

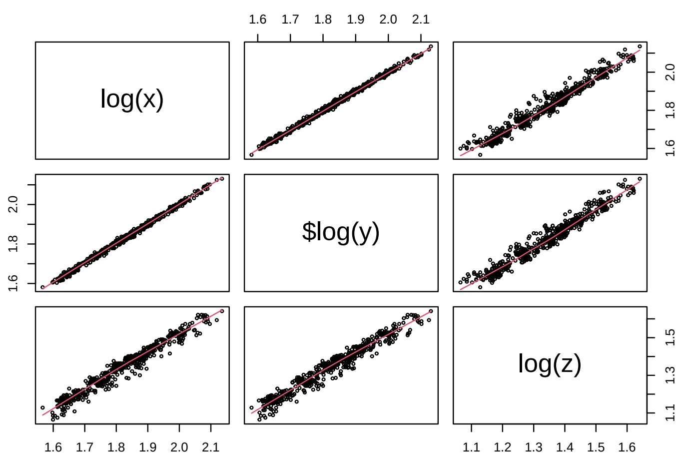

Figure 1: Pairwise scatterplots for $\log(x)$, $\log(y)$ and $\log(z)$.

The large difference between the estimated slope coefficients and the expected values from part (b) are likely due to multi-collinearity. As x, y and z measure the physical dimensions of the same diamond, all variables measure similar information causing the slope coefficients to be unstable. In addition, this may make important variables seem unimportant which is perhaps a reason behind the insufficient evidence to reject the null hypothesis for $\log(x)$ in parts (b) and (c).

To detect multi-collinearity, we use the Variance Inflation Factor (VIF) which calculates the amount a variance of a variable has been increased due to multi-collinearity.

Table 1: VIF values for $\log(x)$, $\log(y)$, $\log(z)$

log(x)	log(y)	log(z)
229.5697	208.493	23.08002

Usually, we consider a VIF > 10 large and thus implying multi-collinearity. From Table 1, we see the VIF for $\log(x)$, $\log(y)$ and $\log(z)$ are greater than 10, although the VIF for $\log(z)$ is considerably smaller, hence implying multi-collinearity between all three predictors.

Furthermore, Figure 1 confirms these predictors are almost perfectly correlated explaining why the individual coefficient estimates in M2 are unstable.

Question 2

a.

b.

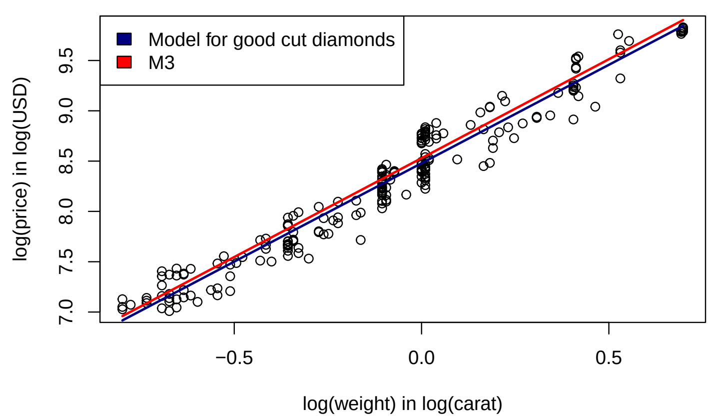

Figure 2: Scatterplot of the logarithm of the price against the logarithm of the weight with fitted curves of the linear model for good cut diamonds and M3 which take observations from good cut and all cut types respectively.

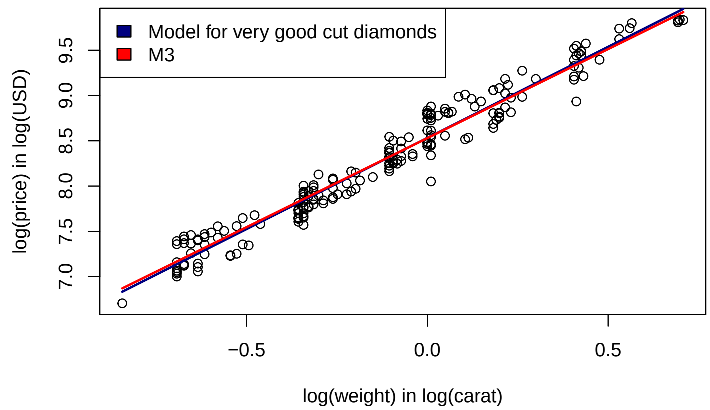

Figure 3: Scatterplot of the logarithm of the price against the logarithm of the weight with fitted curves of the linear model for very good cut diamonds and M3 which take observations from very good cut and all cut types respectively.

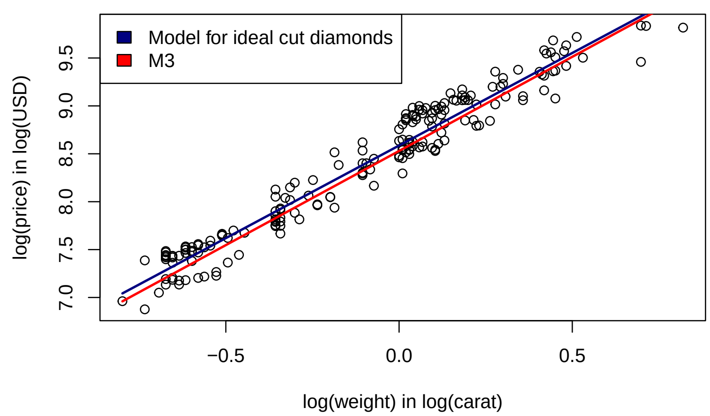

Figure 4: Scatterplot of the logarithm of the price against the logarithm of the weight with fitted curves of the linear model for ideal cut diamonds and M3 which take observations from ideal cut and all cut types respectively.

c.

All three scatterplots show a strong linear relationship between log(weight) and log(price) and hence a linear model is suitable in each case. However, the regression line fitted to each subset is different from the fitted line for M3, particularly in Figure 4. From Figure 2, for good cut diamonds, we see that the subset line lies below the M3 line implying M3 overestimates the price of good cut diamonds on average. From Figure 3, for very good cut diamonds, the fitted lines are almost identical implying M3 estimates the price of very good cut diamonds precisely. Finally, from Figure 4, for ideal cut diamonds, the subset line is above M3 implying M3 systematically underestimates the price of ideal cut diamonds.

Noticeably, the slopes of all four lines are approximately equal which suggests there is little use of adding an interaction between cut and log(weight). However, their intercepts differ significantly across the cut groups which implies adding cut as a main effect is useful, since it would account for the vertical shift in price associated with diamonds of equal weights.

d.

```
Analysis of Variance Table

Model 1: log.price ~ log.weight
Model 2: log.price ~ log.weight + cut
Model 3: log.price ~ log.weight + cut + log.weight:cut
  Res.Df    RSS Df Sum of Sq       F    Pr(>F)
1    605 16.756
2    603 15.526  2   1.23049 23.9468 9.855e-11 ***
3    601 15.441  2   0.08507  1.6556    0.1918
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since M3 is nested within m_{cut}, which in turn is nested within m_{inter}, we use sequential F-tests via ANOVA. For adding cut as a main effect, the **null and alternative hypotheses** are

$$H_0 : \beta_{\text{cut}_1} = \beta_{\text{cut}_2} = 0, \quad \text{versus} \quad H_A : (\beta_{\text{cut}_1}, \beta_{\text{cut}_2}) \neq (0, 0).$$

Since the p-value is $<< 0.01$, at the 1% significance level we reject H_0 and conclude that cut is included in the main effect. Hence, the cut quality of the diamond is a significant predictor of its price after accounting for weight.

For adding the interaction, the **null and alternative hypotheses** are

$$H_0 : \beta_{\text{inter}_1} = \beta_{\text{inter}_2} = 0, \quad \text{versus} \quad H_A : (\beta_{\text{inter}_1}, \beta_{\text{inter}_2}) \neq (0, 0).$$

From row three of the $\Pr(> |t|)$ column in the output, we find the p-value to be $0.192 > 0.01$ (to 3.d.p) there is insufficient evidence to reject the null hypothesis. Hence, the interaction between cut and log(weight) is not needed which is consistent with the conclusions made from the scatter plots in part (c).

e.

```
Call:
lm(formula = log.price ~ cut/log.weight, data = dia)

Residuals:
     Min       1Q   Median       3Q      Max
```

```
-0.50077 -0.11321 -0.01225  0.13273  0.33827

Coefficients:
                        Estimate Std. Error t value Pr(>|t|)
(Intercept)              8.47893    0.01132 749.253  < 2e-16 ***
cutIdeal                 0.10708    0.01648   6.497 1.72e-10 ***
cutVery Good             0.05284    0.01663   3.178  0.00156 **
cutGood:log.weight       1.95629    0.03054  64.062  < 2e-16 ***
cutIdeal:log.weight      1.93300    0.03089  62.577  < 2e-16 ***
cutVery Good:log.weight  2.01222    0.03214  62.615  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1603 on 601 degrees of freedom
Multiple R-squared:  0.9523,    Adjusted R-squared:  0.9519
F-statistic:  2401 on 5 and 601 DF,  p-value: < 2.2e-16
```

The formula cut/ log(weight) is the same as cut + cut : log weight. For each cut group, this fits a model with a separate intercept and slope.

Firstly, let x_{iG}, x_iVG and x_jI as indicator variables for good, very good and ideal cut groups respectively and $j = 1, \ldots, 607$. Using "good cut" as the baseline group, the model equation for the i^{th} observation is:

$$\log(\text{price}_i) = c_G + \alpha_{VG}x_{iVG} + \alpha_I x_{iI} + \beta_G \log(\text{weight}_i)x_{iG} + \beta_{VG}\log(\text{weight}_i)x_{iVG} + \beta_I \log(\text{weight}_i)x_{iI} + \epsilon_i \quad (1)$$

We can interpret the parameters as follows:

1. c_G is the intercept of the "good cut" diamonds which was our baseline group.
2. α_{VG} is the difference between the intercepts of the "good cut" and "very good" cut regression lines.
3. α_I is the difference between the intercepts of the "good cut" and "ideal cut" regression lines.
4. β_G, β_{VG}, β_I are the gradients of log(weight) for good, very good and ideal cut diamonds respectively.
5. Finally, ϵ_i is the error term for the i^{th} observation.

Rounded to 3.d.p, the least squares estimates are: $\hat{c}_G = 8.479, \hat{\alpha}_{VG} = 0.0528, \hat{\alpha}_I = 0.107, \hat{\beta}_G = 1.956, \hat{\beta}_{VG} = 2.012, \hat{\beta}_I = 1.933$.

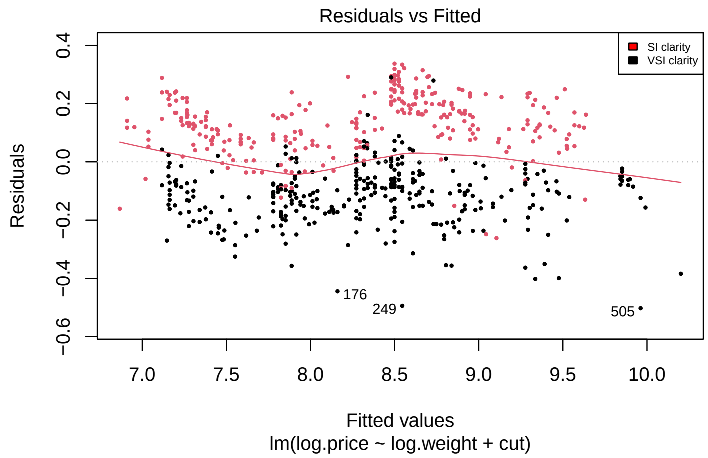

Figure 5: Residuals versus fitted values in log(USD) from M4 overlaid with a Lowess smoother. Red and black data points represent SI and VSI clarity diamonds respectively.

Question 3

a.

From Figure 5, we notice a clear systematic structure in the data by clarity group. Observations with Slightly Imperfect (SI) clarity tend to remain below the zero line while observations with Very Slightly Imperfect (VSI) clarity tend to remain above the zero line across all the fitted values. This suggests M4 overestimates the log(price) for SI diamonds and underestimates log(price) for VSI diamonds. Hence, this systematic pattern strongly suggests adding clarity as a predictor to the model will improve the model.

b.

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Analysis of Variance Table

Model 1: log.price ~ log.weight + cut
Model 2: log.price ~ log.weight + cut + clarity
  Res.Df     RSS Df Sum of Sq      F    Pr(>F)
1    603 15.5260
2    602  6.0558  1    9.4702 941.41 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

As M4 is nested within M5 (obtained by setting $\beta_{\text{clarity}} = 0$ in M5), we use a partial F-test.

The model under H_0 is $\mathbb{E}[\log(\text{price})] = \beta_0 + \beta_1 \log(\text{weight}) + \beta_{\text{cut}}$, that is, for M4, clarity has no effect on price beyond weight and cut. Additionally, the model under H_A is $\mathbb{E}[\log(\text{price})] = \beta_0 + \beta_1 \log(\text{weight}) + \beta_{\text{cut}} + \beta_{\text{clarity}}$, i.e. that is, for M5, clarity has a non-zero effect on price.

Now, we find the algebraic expression for the test statistic. Let

1. D_{M4} and D_{M5} be the residual sums of squares of M4 and M5 respectively,
2. q be the number of additional parameters added when moving from M4 to M5,
3. n is the number of observations, and
4. p is the total number of parameters in M5.

Then, we have:

$$F = \frac{\frac{D_{M4} - D_{M5}}{q}}{\frac{D_{M5}}{n-p}}.$$

Here, we have $n = 607$. As clarity has two levels, SI and VSI, a single indicator variable suffices hence $q = 1$.

M5 has $p = 5$ parameters (intercept, log(weight), two cut dummies and finally one clarity dummy). Therefore, under H_0, the test statistic therefore follows an $F_{(1,602)}$ distribution. Now, using the residual sums of squares from the ANOVA output, we obtain the following, rounding F_{obs} to 1.d.p:

$$F_{\text{obs}} = \frac{\frac{15.526 - 6.056}{1}}{\frac{6.056}{602}} = \frac{9.47}{0.01006} \approx 941.4$$

Since the p-value is less than 0.01, at the 1% significance level we therefore reject the null hypothesis and conclude that clarity is a statistically significant predictor of the price of the diamonds (even after accounting for weight and cut).

In the context of this application, the diamond's clarity grade provides important information regarding the sale price. Hence, a jeweller looking to predict prices accurately should include clarity in the model.

c.

Yes. In this case, the partial F-test could be replaced by a two-sided t-test since $q = 1$. Clarity is a binary variable with two levels, that is SI and VSI, thus only one coefficient is being tested.

When the F-statistic has one degree of freedom in the numerator, the identity $F = t^2$ holds exactly. This implies the partial F-test and the two-sided t-test are equivalent and hence will give the same conclusion.